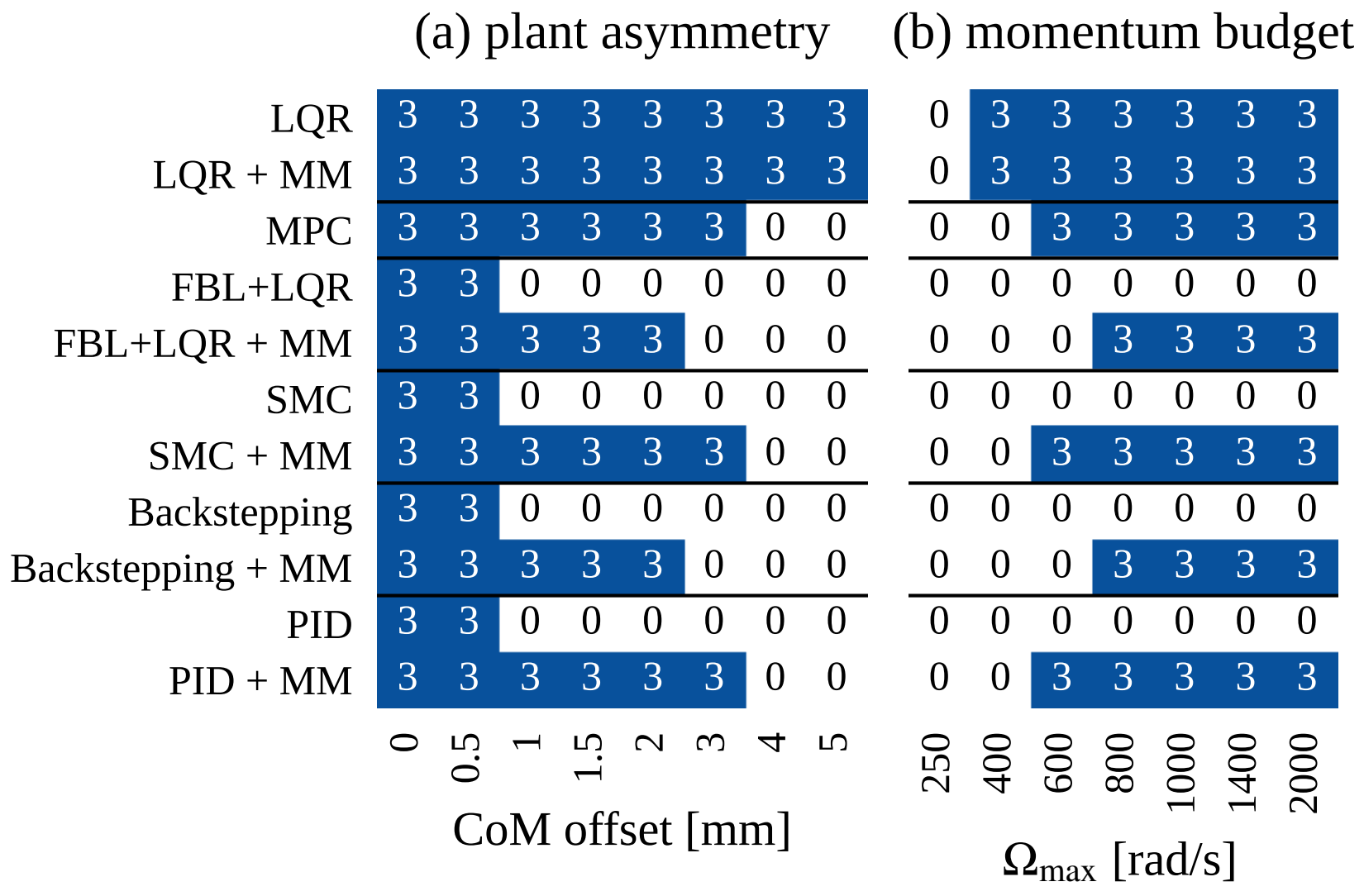
(a) plant asymmetry
(b) momentum budget
LQR
LQR + MM
MPC
FBL+LQR
FBL+LQR + MM
SMC
SMC + MM
Backstepping
Backstepping + MM
PID
PID + MM
3 3 3 3 3 3 3 3
3 3 3 3 3 3 3 3
3 3 3 3 3 3 0 0
3 3 0 0 0 0 0 0
3 3 3 3 3 0 0 0
3 3 0 0 0 0 0 0
3 3 3 3 3 3 0 0
3 3 0 0 0 0 0 0
3 3 3 3 3 0 0 0
3 3 0 0 0 0 0 0
3 3 3 3 3 3 0 0
0 3 3 3 3 3 3
0 3 3 3 3 3 3
0 0 3 3 3 3 3
0 0 0 0 0 0 0
0 0 0 3 3 3 3
0 0 0 0 0 0 0
0 0 3 3 3 3 3
0 0 0 0 0 0 0
0 0 0 3 3 3 3
0 0 0 0 0 0 0
0 0 3 3 3 3 3
0
0.5
1
1.5
2
3
4
5
CoM offset [mm]
250
400
600
800
1000
1400
2000
Ωmax [rad/s]